College of Engineering, Construction and Living Sciences
Bachelor of Information Technology
ID607001: Introductory Application Development Concepts
Level 6, Credits 15

Project

Assessment Overview

In this **individual** assessment, you will develop two web applications using **Express** and **SvelteKit**.

Learning Outcome

At the successful completion of this course, learners will be able to:

1. Design and build secure applications with dynamic database functionality following an appropriate software development methodology.

Assessments

Assessment	Weighting	Due Date	Learning Outcome
Practical	20%	Sunday 7th September at 11.59 PM	1
Project	80%	Sunday 9th November at 11.59 PM	1

Conditions of Assessment

You will complete this assessment mostly during your learner-managed time. However, there will be time during class to discuss the requirements and your progress on this assessment. This assessment will need to be completed by **Sunday 9th November** at **11.59 PM**.

Pass Criteria

This assessment is criterion-referenced (CRA) with a cumulative pass mark of **50%** across all assessments in **ID607001: Introductory Application Development Concepts**.

Submission

You **must** submit all application files via **GitHub Classroom**. Here is the URL to the repository you will use for your submission – https://classroom.github.com/a/8sCyquQ_. If you do not have not one, create a **.gitignore** and add the ignored files in this resource - https://raw.githubusercontent.com/github/gitignore/main/Node.gitignore.

Create a branch called **project**. The latest application files in the **project** branch will be used to mark against the marking rubric. Please test before you submit. Partial marks **will not** be given for incomplete functionality. Late submissions will incur a **10% penalty per day**, rolling over at **12.00 AM**.

Authenticity

All parts of your submitted assessment **must** be completely your work. Do your best to complete this assessment without using AI tools. You need to demonstrate to the course lecturer that you can meet the learning outcome for this assessment.

Learning to use AI tools is an important skill. While AI tools are powerful, you **must** be aware of the following:

- If you provide an AI tool with a prompt that is not refined enough, it may generate a not-so-useful response
- Do not trust the AI tool's responses blindly. You **must** still use your judgement and may need to do additional research to determine if the response is correct
- Acknowledge what AI tool you have used. In the assessment's repository **README.md** file, please include what prompt(s) you provided to the AI tool and how you used the response(s) to help you with your work

It also applies to code snippets retrieved from **StackOverflow** and **GitHub**.

Failure to do this may result in a mark of **zero** for this assessment.

Policy on Submissions, Extensions, Resubmissions and Resits

The school's process concerning submissions, extensions, resubmissions and resits complies with **Otago Polytechnic** policies. Learners can view policies on the **Otago Polytechnic** website located at https://www.op.ac.nz/about-us/governance-and-management/policies.

Extensions

Familiarise yourself with the assessment due date. Extensions will **only** be granted if you are unable to complete the assessment by the due date because of **unforeseen circumstances outside your control**. The length of the extension granted will depend on the circumstances and **must** be negotiated with the course lecturer before the assessment due date. A medical certificate or support letter may be needed. Extensions will not be granted on the due date and for poor time management or pressure of other assessments.

Resits

Resits and reassessments are not applicable in **ID607001: Introductory Application Development Concepts**.

Instructions

Functionality - Learning Outcome 1 (25%)

- Backend application with the following functionality:
  - Five **models** with a minimum of four **fields**.
  - One model should have an enum field.
  - One **one-to-many** relationship and one **many-to-many** relationship.
  - **CRUD** operations for each model.
  - Validation on **Create** and **Update** operations.
  - **Filter**, **sort** and **paginate** using **query parameters**.
  - Allow users to register, login and logout.
  - Role-based access control:
    - An **admin user** can perform **CRUD** operations on all **models**.
    - A **normal user** can perform **Read all** and **Read by UUID** operations on all **models**.
  - Content negotiation to support **JSON** format.
  - Seed script to populate the database with initial data.
  - API tests covering **CRUD** operations for each model.
  - Deploy to **Render**.
- Frontend application with the following functionality:
  - Consume the backend application on **Render** to perform **CRUD** operations based on role.
  - Styled using **Bootstrap**.
  - Authentication with secure token handling.
  - Display data with support for **filtering**, **sorting** and **paging**.
  - Forms with validation for **Create** and **Update** operations.
  - Render UI elements and actions based on role.
  - Deploy to **Vercel**.

Code Quality and Best Practices - Learning Outcome 1 (40%)

- Write clean, readable and maintainable code.
- Use modular, reusable components and avoid code duplication.
- Keep code simple and easy to understand.
- Optimise performance by managing resources efficiently.

Version Control - Learning Outcome 1 (5%)

- Regular commits are made throughout development, demonstrating meaningful progress.
- Commit messages are clear and descriptive.
- The repository shows a clean and logical commit history that reflects the development process of the applications.

Reflection - Learning Outcome 1 (5%)

- Write a short reflection (300-500 words) on your development process and learning throughout the project, including:
  - Challenges you faced and how you overcame them.
  - What you would do differently in future projects.
  - New skills, tools or practices you learned during the project.

Presentation - Learning Outcome 1 (5%)

- Record a 5-10 minute screen recording of your applications.
- Demonstrate the backend and frontend features.
- Make sure the video is clear and easy to follow.
- Submit a link to the presentation in your repository's **README.md** file.